KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4051 | 4 Credits: 4 | Program Core | Source-grounded

# Heat Power Engineering

 To get elementary knowledge about thermodynamic processes, cycles, fundamental

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4051 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 4051.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

## Official course information

**Course information**

| Item | Official information |
|---|---|
| Programme | Diploma in Automobile Engineering |
| Course code | 4051 |
| Course title | Heat Power Engineering |
| Semester | 4 Credits: 4 |
| Course category | Program Core |
| Periods per week | 4 (L:3, T:1, P:0) Periods per semester: 60 |

| Item | Official information |
|---|---|
| Periods per semester | 60 |
| Credits | 4 |
| Prerequisite | See the official syllabus PDF |
| Assessment | Use the official assessment section reproduced below |

### Modules

4 official module sections detected.

### Topic entries

20 source-aligned topic entries retained.

### Study mode

Theory and application emphasis.

STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

-  To get elementary knowledge about thermodynamic processes, cycles, fundamental
- laws of thermodynamics.
-  To understand the laws of conversion of heat energy into work or power and its
- applications in automobile industries.
-  To deal with thermodynamic systems such as heat engines, heat exchangers
- (radiators), automobile air conditioning, compressors etc.
-  To understand the terminologies used in the testing of engine performances.

**Course outcomes**

| CO | Official outcome | Study level |
|---|---|---|
| CO1 | 15 Applying in thermodynamics Select air standard efficiency of Carnot cycle, Otto | See official syllabus |
| CO2 | 14 Applying cycle and Diesel cycles. | See official syllabus |
| CO3 | Solve the performance parameters of engines. 15 Applying Make use of the principles of heat transfer and | See official syllabus |
| CO4 | 14 Applying working of air compressors. | See official syllabus |

**CO–PO mapping evidence**

| Row | Extracted official mapping |
|---|---|
| CO1 | CO1 3 |
| CO2 | CO2 3 |
| CO3 | CO3 3 |

| Row | Extracted official mapping |
|---|---|
| CO4 | CO4 3 |

COVERAGE AUDIT

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

**Module coverage**

| Module | Official heading | Topic entries | Hours |
|---|---|---|---|
| Module 1 | Choose the basic terms, laws and types of processes in thermodynamics. | 4 | As specified |
| Module 2 | Select air standard efficiency of Carnot cycle, Otto cycle and Diesel cycles | 6 | As specified |
| Module 3 | Solve the performance parameters of engines. | 5 | As specified |
| Module 4 | compressors. | 5 | As specified |

MODULE 1

# Choose the basic terms, laws and types of processes in thermodynamics.

This module is presented from the official syllabus scope for Heat Power Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Choose the basic terms, laws and types of processes in thermodynamics.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

**Define basic terms of thermodynamics 3 Remembering Explain First, Second, Third and Zeroth Laws M10.2 1 Understanding of thermodynamics. Summarize the basic concepts of entropy,**

Define basic terms of thermodynamics 3 Remembering Explain First, Second, Third and Zeroth Laws M10.2 1 Understanding of thermodynamics. Summarize the basic concepts of entropy, is an official topic in Module 1 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Define basic terms of thermodynamics 3 Remembering Explain First, Second, Third and Zeroth Laws M10.2 1 Understanding of thermodynamics. Summarize the basic concepts of entropy, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, define basic terms of thermodynamics 3 remembering explain first, second, third and zeroth laws m10.2 1 understanding of thermodynamics. summarize the basic concepts of entropy, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Define basic terms of thermodynamics 3 Remembering Explain First, Second, Third and Zeroth Laws M10.2 1 Understanding of thermodynamics. Summarize the basic concepts of entropy, means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Define basic terms of thermodynamics 3 Remembering Explain First, Second, Third and Zeroth Laws M10.2 1 Understanding of thermodynamics. Summarize the basic concepts of entropy, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 2 Understanding enthalpy and internal energy. Make use of various types of thermodynamic

2 Understanding enthalpy and internal energy. Make use of various types of thermodynamic is an official topic in Module 1 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding enthalpy and internal energy. Make use of various types of thermodynamic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding enthalpy and internal energy. make use of various types of thermodynamic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding enthalpy and internal energy. Make use of various types of thermodynamic means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 2 Understanding enthalpy and internal energy. Make use of various types of thermodynamic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 6 Applying processes. Solve problems on different types of

6 Applying processes. Solve problems on different types of is an official topic in Module 1 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 6 Applying processes. Solve problems on different types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 6 applying processes. solve problems on different types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 6 Applying processes. Solve problems on different types of means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 6 Applying processes. Solve problems on different types of □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

**3 Applying thermodynamic processes. Fundamentals of thermodynamics and thermodynamic processes Classification of thermodynamics, Application areas of thermodynamics, Dimensional homogeneity, Basics of thermodynamics - thermodynamic system, surroundings, boundary, types of system, properties of a system, state, thermodynamic equilibrium, Ideal gas equation of state - Derivation, pressure and its units, temperature and its units, Zeroth law of thermodynamics, Specific heat at constant pressure and constant volume, relationship between specific heats and gas constant, Internal Energy, flow energy and moving boundary work, First law of thermodynamics - Enthalpy, Second law of thermodynamics, Entropy-Third law of thermodynamics. Processes and cycle. Pressure volume diagram, Temperature entropy diagram, Types of processes - Derivations on PVT relationship, heat transfer, work transfer, change in entropy, enthalpy and internal energy of isochoric, isobaric, isothermal and adiabatic processes. Simple problems to find change in internal energy, heat transfer, work transfer etc.**

3 Applying thermodynamic processes. Fundamentals of thermodynamics and thermodynamic processes Classification of thermodynamics, Application areas of thermodynamics, Dimensional homogeneity, Basics of thermodynamics - thermodynamic system, surroundings, boundary, types of system, properties of a system, state, thermodynamic equilibrium, Ideal gas equation of state - Derivation, pressure and its units, temperature and its units, Zeroth law of thermodynamics, Specific heat at constant pressure and constant volume, relationship between specific heats and gas constant, Internal Energy, flow energy and moving boundary work, First law of thermodynamics - Enthalpy, Second law of thermodynamics, Entropy-Third law of thermodynamics. Processes and cycle. Pressure volume diagram, Temperature entropy diagram, Types of processes - Derivations on PVT relationship, heat transfer, work transfer, change in entropy, enthalpy and internal energy of isochoric, isobaric, isothermal and adiabatic processes. Simple problems to find change in internal energy, heat transfer, work transfer etc. is an official topic in Module 1 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

## Study points

- Define or identify 3 Applying thermodynamic processes. Fundamentals of thermodynamics and thermodynamic processes Classification of thermodynamics, Application areas of thermodynamics, Dimensional homogeneity, Basics of thermodynamics - thermodynamic system, surroundings, boundary, types of system, properties of a system, state, thermodynamic equilibrium, Ideal gas equation of state - Derivation, pressure and its units, temperature and its units, Zeroth law of thermodynamics, Specific heat at constant pressure and constant volume, relationship between specific heats and gas constant, Internal Energy, flow energy and moving boundary work, First law of thermodynamics - Enthalpy, Second law of thermodynamics, Entropy-Third law of thermodynamics. Processes and cycle. Pressure volume diagram, Temperature entropy diagram, Types of processes - Derivations on PVT relationship, heat transfer, work transfer, change in entropy, enthalpy and internal energy of isochoric, isobaric, isothermal and adiabatic processes. Simple problems to find change in internal energy, heat transfer, work transfer etc. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 3 applying thermodynamic processes. fundamentals of thermodynamics and thermodynamic processes classification of thermodynamics, application areas of thermodynamics, dimensional homogeneity, basics of thermodynamics - thermodynamic system, surroundings, boundary, types of system, properties of a system, state, thermodynamic equilibrium, ideal gas equation of state - derivation, pressure and its units, temperature and its units, zeroth law of thermodynamics, specific heat at constant pressure and constant volume, relationship between specific heats and gas constant, internal energy, flow energy and moving boundary work, first law of thermodynamics - enthalpy, second law of thermodynamics, entropy-third law of thermodynamics. processes and cycle. pressure volume diagram, temperature entropy diagram, types of processes - derivations on pvt relationship, heat transfer, work transfer, change in entropy, enthalpy and internal energy of isochoric, isobaric, isothermal and adiabatic processes. simple problems to find change in internal energy, heat transfer, work transfer etc. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Applying thermodynamic processes. Fundamentals of thermodynamics and thermodynamic processes Classification of thermodynamics, Application areas of thermodynamics, Dimensional homogeneity, Basics of thermodynamics - thermodynamic system, surroundings, boundary, types of system, properties of a system, state, thermodynamic equilibrium, Ideal gas equation of state - Derivation, pressure and its units, temperature and its units, Zeroth law of thermodynamics, Specific heat at constant pressure and constant volume, relationship between specific heats and gas constant, Internal Energy, flow energy and moving boundary work, First law of thermodynamics - Enthalpy, Second law of thermodynamics, Entropy-Third law of thermodynamics. Processes and cycle. Pressure volume diagram, Temperature entropy diagram, Types of processes - Derivations on PVT relationship, heat transfer, work transfer, change in entropy, enthalpy and internal energy of isochoric, isobaric, isothermal and adiabatic processes. Simple problems to find change in internal energy, heat transfer, work transfer etc. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 3 Applying thermodynamic processes. Fundamentals of thermodynamics and thermodynamic processes Classification of thermodynamics, Application areas of thermodynamics, Dimensional homogeneity, Basics of thermodynamics - thermodynamic system, surroundings, boundary, types of system, properties of a system, state, thermodynamic equilibrium, Ideal gas equation of state - Derivation, pressure and its units, temperature and its units, Zeroth law of thermodynamics, Specific heat at constant pressure and constant volume, relationship between specific heats and gas constant, Internal Energy, flow energy and moving boundary work, First law of thermodynamics - Enthalpy, Second law of thermodynamics, Entropy-Third law of thermodynamics. Processes and cycle. Pressure volume diagram, Temperature entropy diagram, Types of processes - Derivations on PVT relationship, heat transfer, work transfer, change in entropy, enthalpy and internal energy of isochoric, isobaric, isothermal and adiabatic processes. Simple problems to find change in internal energy, heat transfer, work transfer etc. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Choose the basic terms, laws and types of processes in thermodynamics.

  M1.01      Define basic terms of thermodynamics                   3
Remembering
             Explain First, Second, Third and Zeroth Laws
  M10.2                                                             1
Understanding
             of thermodynamics.
             Summarize the basic concepts of entropy,
  M1.03                                                             2
Understanding
             enthalpy and internal energy.
             Make use of various types of thermodynamic
  M1.04                                                             6
Applying
             processes.
             Solve problems on different types of
  M1.05                                                             3
Applying
             thermodynamic processes.

Contents:
Fundamentals of thermodynamics and thermodynamic processes
Classification of thermodynamics, Application areas of thermodynamics,
Dimensional
homogeneity, Basics of thermodynamics - thermodynamic system, surroundings,
boundary, types of system, properties of a system, state, thermodynamic
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

# Select air standard efficiency of Carnot cycle, Otto cycle and Diesel cycles

This module is presented from the official syllabus scope for Heat Power Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

### Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Select air standard efficiency of Carnot cycle, Otto cycle and Diesel cycles
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

## thermodynamic cycles. Classify and compare 2 Understanding, thermodynamic cycles. Define available energy, theoretical thermal

thermodynamic cycles. Classify and compare 2 Understanding, thermodynamic cycles. Define available energy, theoretical thermal is an official topic in Module 2 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify thermodynamic cycles. Classify and compare 2 Understanding, thermodynamic cycles. Define available energy, theoretical thermal using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, thermodynamic cycles. classify and compare 2 understanding, thermodynamic cycles. define available energy, theoretical thermal should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What thermodynamic cycles. Classify and compare 2 Understanding, thermodynamic cycles. Define available energy, theoretical thermal means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] thermodynamic cycles. Classify and compare 2 Understanding, thermodynamic cycles. Define available energy, theoretical thermal [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 2 Remembering, efficiency / air standard efficiency. Summarize the theoretical thermal efficiency

2 Remembering, efficiency / air standard efficiency. Summarize the theoretical thermal efficiency is an official topic in Module 2 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Remembering, efficiency / air standard efficiency. Summarize the theoretical thermal efficiency using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 remembering, efficiency / air standard efficiency. summarize the theoretical thermal efficiency should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Remembering, efficiency / air standard efficiency. Summarize the theoretical thermal efficiency means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] 2 Remembering, efficiency / air standard efficiency. Summarize the theoretical thermal efficiency [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 2 Understanding of Carnot cycle.

2 Understanding of Carnot cycle. is an official topic in Module 2 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding of Carnot cycle. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding of carnot cycle. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 2 Understanding of Carnot cycle. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] 2 Understanding of Carnot cycle. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Solve simple problems on Carnot cycle. 2 Applying Explain theoretical thermal efficiency of Otto

Solve simple problems on Carnot cycle. 2 Applying Explain theoretical thermal efficiency of Otto is an official topic in Module 2 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Solve simple problems on Carnot cycle. 2 Applying Explain theoretical thermal efficiency of Otto using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, solve simple problems on carnot cycle. 2 applying explain theoretical thermal efficiency of otto should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Solve simple problems on Carnot cycle. 2 Applying Explain theoretical thermal efficiency of Otto means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Solve simple problems on Carnot cycle. 2 Applying Explain theoretical thermal efficiency of Otto □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### 3 Understanding, cycle and Diesel Cycle. Solve problems on Otto cycle and Diesel

3 Understanding, cycle and Diesel Cycle. Solve problems on Otto cycle and Diesel is an official topic in Module 2 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding, cycle and Diesel Cycle. Solve problems on Otto cycle and Diesel using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding, cycle and diesel cycle. solve problems on otto cycle and diesel should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding, cycle and Diesel Cycle. Solve problems on Otto cycle and Diesel means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 3 Understanding, cycle and Diesel Cycle. Solve problems on Otto cycle and Diesel □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**3 Applying cycle. Air standard cycles. Assumptions made in thermodynamic cycles, Classification of thermodynamic cycles. Available energy, theoretical thermal efficiency, Air standard efficiency. Derivation of theoretical thermal efficiency of Carnot cycle, Otto cycle and Diesel cycle -Simple problems to find out air standard efficiency.**

3 Applying cycle. Air standard cycles. Assumptions made in thermodynamic cycles, Classification of thermodynamic cycles. Available energy, theoretical thermal efficiency, Air standard efficiency. Derivation of theoretical thermal efficiency of Carnot cycle, Otto cycle and Diesel cycle -Simple problems to find out air standard efficiency. is an official topic in Module 2 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Applying cycle. Air standard cycles. Assumptions made in thermodynamic cycles, Classification of thermodynamic cycles. Available energy, theoretical thermal efficiency, Air standard efficiency. Derivation of theoretical thermal efficiency of Carnot cycle, Otto cycle and Diesel cycle -Simple problems to find out air standard efficiency. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 applying cycle. air standard cycles. assumptions made in thermodynamic cycles, classification of thermodynamic cycles. available energy, theoretical thermal efficiency, air standard efficiency. derivation of theoretical thermal efficiency of carnot cycle, otto cycle and diesel cycle -simple problems to find out air standard efficiency. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Applying cycle. Air standard cycles. Assumptions made in thermodynamic cycles, Classification of thermodynamic cycles. Available energy, theoretical thermal efficiency, Air standard efficiency. Derivation of theoretical thermal efficiency of Carnot cycle, Otto cycle and Diesel cycle -Simple problems to find out air standard efficiency. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 3 Applying cycle. Air standard cycles. Assumptions made in thermodynamic cycles, Classification of thermodynamic cycles. Available energy, theoretical thermal efficiency, Air standard efficiency. Derivation of theoretical thermal efficiency of Carnot cycle, Otto cycle and Diesel cycle -Simple problems to find out air standard efficiency. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Select air standard efficiency of Carnot cycle, Otto cycle and Diesel cycles
            Summarize the assumptions made in
  M2.01     thermodynamic cycles. Classify and compare         2
Understanding,
            thermodynamic cycles.
            Define available energy, theoretical thermal
  M2.02                                                        2
Remembering,
            efficiency / air standard efficiency.
            Summarize the theoretical thermal efficiency
  M2.03                                                        2
Understanding
            of Carnot cycle.

  M2.04     Solve simple problems on Carnot cycle.             2
Applying


            Explain theoretical thermal efficiency of Otto
  M2.05                                                        3
Understanding,
            cycle and Diesel Cycle.
            Solve problems on Otto cycle and Diesel
  M2.06                                                        3
Applying
            cycle.
            Series test -I                                     1
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

# Solve the performance parameters of engines.

This module is presented from the official syllabus scope for Heat Power Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Solve the performance parameters of engines.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

**efficiency, brake thermal efficiency, 6 Understanding volumetric efficiency, Mechanical efficiency, Mean effective pressures, and specific fuel consumption. Solve the problems related to testing of**

efficiency, brake thermal efficiency, 6 Understanding volumetric efficiency, Mechanical efficiency, Mean effective pressures, and specific fuel consumption. Solve the problems related to testing of is an official topic in Module 3 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify efficiency, brake thermal efficiency, 6 Understanding volumetric efficiency, Mechanical efficiency, Mean effective pressures, and specific fuel consumption. Solve the problems related to testing of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, efficiency, brake thermal efficiency, 6 understanding volumetric efficiency, mechanical efficiency, mean effective pressures, and specific fuel consumption. solve the problems related to testing of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What efficiency, brake thermal efficiency, 6 Understanding volumetric efficiency, Mechanical efficiency, Mean effective pressures, and specific fuel consumption. Solve the problems related to testing of means in the official course context |
| Study lens | Principle → components or stages → result → application |

| Aspect | What to prepare |
| --- | --- |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ efficiency, brake thermal efficiency, 6 Understanding volumetric efficiency, Mechanical efficiency, Mean effective pressures, and specific fuel consumption. Solve the problems related to testing of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 4 Applying Internal combustion engines. Summarize how Morse test is conducted in

4 Applying Internal combustion engines. Summarize how Morse test is conducted in is an official topic in Module 3 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Applying Internal combustion engines. Summarize how Morse test is conducted in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 applying internal combustion engines. summarize how morse test is conducted in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 4 Applying Internal combustion engines. Summarize how Morse test is conducted in means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] 4 Applying Internal combustion engines. Summarize how Morse test is conducted in [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 1 Understanding multi cylinder engines.

1 Understanding multi cylinder engines. is an official topic in Module 3 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding multi cylinder engines. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding multi cylinder engines. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 1 Understanding multi cylinder engines. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ 1 Understanding multi cylinder engines. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## Explain how heat balance sheets are prepared. 1 Understanding Solve the problems on Morse test and Heat

Explain how heat balance sheets are prepared. 1 Understanding Solve the problems on Morse test and Heat is an official topic in Module 3 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain how heat balance sheets are prepared. 1 Understanding Solve the problems on Morse test and Heat using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain how heat balance sheets are prepared. 1 understanding solve the problems on morse test and heat should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Explain how heat balance sheets are prepared. 1 Understanding Solve the problems on Morse test and Heat means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Explain how heat balance sheets are prepared. 1 Understanding Solve the problems on Morse test and Heat □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**3 Applying balance sheet Performance parameters - indicated power, brake power, friction power, indicated thermal efficiency, brake thermal efficiency, volumetric efficiency, Mechanical efficiency, mean effective pressures and specific fuel consumption - Simple problems to find out various efficiencies. Morse test and preparation of heat balance sheet. Simple problems. Make use of the principles of heat transfer and working of air**

3 Applying balance sheet Performance parameters - indicated power, brake power, friction power, indicated thermal efficiency, brake thermal efficiency, volumetric efficiency, Mechanical efficiency, mean effective pressures and specific fuel consumption - Simple problems to find out various efficiencies. Morse test and preparation of heat balance sheet. Simple problems. Make use of the principles of heat transfer and working of air is an official topic in Module 3 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Applying balance sheet Performance parameters - indicated power, brake power, friction power, indicated thermal efficiency, brake thermal efficiency, volumetric efficiency, Mechanical efficiency, mean effective pressures and specific fuel consumption - Simple problems to find out various efficiencies. Morse test and preparation of heat balance sheet. Simple problems. Make use of the principles of heat transfer and working of air using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 applying balance sheet performance parameters - indicated power, brake power, friction power, indicated thermal efficiency, brake thermal efficiency, volumetric efficiency, mechanical efficiency, mean effective pressures and specific fuel consumption - simple problems to find out various efficiencies. morse test and preparation of heat balance sheet. simple problems. make use of the principles of heat transfer and working of air should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Applying balance sheet Performance parameters - indicated power, brake power, friction power, indicated thermal efficiency, brake thermal efficiency, volumetric efficiency, Mechanical efficiency, mean effective pressures and specific fuel consumption - Simple problems to find out various efficiencies. Morse test and preparation of heat balance sheet. Simple problems. Make use of the principles of heat transfer and working of air means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ 3 Applying balance sheet Performance parameters - indicated power, brake power, friction power, indicated thermal efficiency, brake thermal efficiency, volumetric efficiency, Mechanical efficiency, mean effective pressures and specific fuel consumption - Simple problems to find out various efficiencies. Morse test and preparation of heat balance sheet. Simple problems. Make use of the principles of heat transfer and working of air □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Solve the performance parameters of engines.
            Outline the Performance parameters of I.C.
            engines such as indicated power, brake
            power, friction power, indicated thermal
  M3.01     efficiency, brake thermal efficiency,              6
Understanding
            volumetric efficiency, Mechanical efficiency,
            Mean effective pressures, and specific fuel
            consumption.
            Solve the problems related to testing of
  M3.02                                                        4
Applying
            Internal combustion engines.
            Summarize how Morse test is conducted in
  M3.03                                                        1
Understanding
            multi cylinder engines.
  M3.04     Explain how heat balance sheets are prepared.      1
Understanding
            Solve the problems on Morse test and Heat
  M3.05                                                        3
Applying
            balance sheet
Contents:
Performance parameters - indicated power, brake power, friction power, indicated
thermal
efficiency, brake thermal efficiency, volumetric efficiency, Mechanical
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

## compressors.

This module is presented from the official syllabus scope for Heat Power Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: compressors.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## Identify the fields of application of heat 4 Applying transfer. Compare three modes of heat transfer and

Identify the fields of application of heat 4 Applying transfer. Compare three modes of heat transfer and is an official topic in Module 4 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Identify the fields of application of heat 4 Applying transfer. Compare three modes of heat transfer and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, identify the fields of application of heat 4 applying transfer. compare three modes of heat transfer and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Identify the fields of application of heat 4 Applying transfer. Compare three modes of heat transfer and means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Identify the fields of application of heat 4 Applying transfer. Compare three modes of heat transfer and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## their rate equations. Illustrate two types of convective mode of

their rate equations. Illustrate two types of convective mode of is an official topic in Module 4 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify their rate equations. Illustrate two types of convective mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, their rate equations. illustrate two types of convective mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What their rate equations. Illustrate two types of convective mode of means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ their rate equations. Illustrate two types of convective mode of □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## 2 Understanding heat transfer- natural and forced. Define air compressor, explain uses of

2 Understanding heat transfer- natural and forced. Define air compressor, explain uses of is an official topic in Module 4 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding heat transfer- natural and forced. Define air compressor, explain uses of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding heat transfer- natural and forced. define air compressor, explain uses of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding heat transfer- natural and forced. Define air compressor, explain uses of means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-[illegible] 2 Understanding heat transfer- natural and forced. Define air compressor, explain uses of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## 3 Understanding compressed air and classify air compressors. Illustrate the working of single stage and

3 Understanding compressed air and classify air compressors. Illustrate the working of single stage and is an official topic in Module 4 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding compressed air and classify air compressors. Illustrate the working of single stage and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding compressed air and classify air compressors. illustrate the working of single stage and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding compressed air and classify air compressors. Illustrate the working of single stage and means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ 3 Understanding compressed air and classify air compressors. Illustrate the working of single stage and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**multistage air compressors. Explain the merits 4 Understanding and demerits of multistage compression. Text / Reference: T/R Book Title/Author T1 Thermal Engineering R.S.Khurmi S Chand & Co Basic and applied thermodynamics P.K.Nag Tata MCgraw-Hill T2 Niranjan.Murthy Sapna Publications T3 I C Engines V. Ganeshan Tata MCgraw-Hill R. C. Patel and Karmchandani.C .J. - Elements of Heat engines – Acharya R1 book R2 Lakshminarayana - Thermal engineering Vol.-I R3 N.K.Giri - Automobile mechanics – Khanna publishers R. Rudra Moorthy and K. Mayil Swamy - Heat and Mass transfer- pearson R4 education india R5 J G Giles-Vehicle operations and testing R6 V. Ganesan - Internal combustion engines – Tata McGraw-Hill S. P. Sen - Internal combustion engines theory and practice – Khanna R7 publishers R8 Mathur and Sharma - Internal combustion engines – Dhanpat Rai R9 V.L. Malaev - IC Engines – McGraw-Hill R10 P.M. Heldt - IC Engines R11 Ramalingam - Thermal engineering - New age international B B Ghosh - A Text book of Engineering Thermodynamics – Vikas R12 publishing Online Resources: Sl. No Website Link https://www.toppr.com/guides/chemistry/thermodynamics/applications-of- 1 thermodynamics/ 2 https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch21/chemical.php 3 https://en.wikipedia.org/wiki/Thermodynamic_process http://www.thermodynamicsheatengines.com/HeatEnginesVol%202%20Chapt 4 er%207%20RS.pdf https://www.academia.edu/15057433/CHAPTER_8_Testing_of_I.C.Engines,h 5 ttps://www.mechanicalbooster.com/2015/08/difference-between-otto-cycle-and-diesel-cycle.html 6 https://engineeringinsider.org/heat-transfer-conduction-convection-radiation/ https://www.brighthubengineering.com/hvac/63725-effects-of-a-multi-stage- 7 air-compressor/**

multistage air compressors. Explain the merits 4 Understanding and demerits of multistage compression. Text / Reference: T/R Book Title/Author T1 Thermal Engineering R.S.Khurmi S Chand & Co Basic and applied thermodynamics P.K.Nag Tata MCgraw-Hill T2 Niranjan.Murthy Sapna Publications T3 I C Engines V. Ganeshan Tata MCgraw-Hill R. C. Patel and Karmchandani.C .J. - Elements of Heat engines – Acharya R1 book R2 Lakshminarayana - Thermal engineering Vol.-I R3 N.K.Giri - Automobile mechanics – Khanna publishers R. Rudra Moorthy and K. Mayil Swamy - Heat and Mass transfer- pearson R4 education india R5 J G Giles-Vehicle operations and testing R6 V. Ganesan - Internal combustion engines – Tata McGraw-Hill S. P. Sen - Internal combustion engines theory and practice – Khanna R7 publishers R8 Mathur and Sharma - Internal combustion engines – Dhanpat Rai R9 V.L. Malaev - IC Engines – McGraw-Hill R10 P.M. Heldt - IC Engines R11 Ramalingam - Thermal engineering - New age international B B Ghosh - A

Text book of Engineering Thermodynamics – Vikas R12 publishing Online Resources: Sl. No Website Link https://www.toppr.com/guides/chemistry/thermodynamics/applications-of- 1 thermodynamics/ 2 https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch21/chemical.php 3 https://en.wikipedia.org/wiki/Thermodynamic_process http://www.thermodynamicsheatengines.com/HeatEnginesVol%202%20Chapt 4 er%207%20RS.pdf https://www.academia.edu/15057433/CHAPTER_8_Testing_of_I.C.Engines,h 5 ttps://www.mechanicalbooster.com/2015/08/difference-between-otto-cycle- and-diesel-cycle.html 6 https://engineeringinsider.org/heat-transfer-conduction-convection-radiation/ https://www.brighthubengineering.com/hvac/63725-effects-of-a-multi-stage- 7 air-compressor/ is an official topic in Module 4 of Heat Power Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify multistage air compressors. Explain the merits 4 Understanding and demerits of multistage compression. Text / Reference: T/R Book Title/Author T1 Thermal Engineering R.S.Khurmi S Chand & Co Basic and applied thermodynamics P.K.Nag Tata MCgraw-Hill T2 Niranjan.Murthy Sapna Publications T3 I C Engines V. Ganeshan Tata MCgraw-Hill R. C. Patel and Karmchandani.C .J. - Elements of Heat engines – Acharya R1 book R2 Lakshminarayana - Thermal engineering Vol.-I R3 N.K.Giri - Automobile mechanics – Khanna publishers R. Rudra Moorthy and K. Mayil Swamy - Heat and Mass transfer- pearson R4 education india R5 J G Giles- Vehicle operations and testing R6 V. Ganesan - Internal combustion engines – Tata McGraw-Hill S. P. Sen - Internal combustion engines theory and practice – Khanna R7 publishers R8 Mathur and Sharma - Internal combustion engines – Dhanpat Rai R9 V.L. Malaev - IC Engines – McGraw-Hill R10 P.M. Heldt - IC Engines R11 Ramalingam - Thermal engineering - New age international B B Ghosh - A Text book of Engineering Thermodynamics – Vikas R12 publishing Online Resources: Sl. No Website Link https://www.toppr.com/guides/chemistry/thermodynamics/applications-of- 1 thermodynamics/ 2 https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch21/chemical.php 3 https://en.wikipedia.org/wiki/Thermodynamic_process http://www.thermodynamicsheatengines.com/HeatEnginesVol%202%20Chapt 4 er%207%20RS.pdf https://www.academia.edu/15057433/CHAPTER_8_Testing_of_I.C.Engines,h 5 ttps://www.mechanicalbooster.com/2015/08/difference-between-otto-cycle- and-diesel-cycle.html 6 https://engineeringinsider.org/heat-transfer-conduction-convection-radiation/ https://www.brighthubengineering.com/hvac/63725-effects-of-a-multi-stage- 7 air-compressor/ using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, multistage air compressors. explain the merits 4 understanding and demerits of multistage compression. text / reference: t/r book title/author t1 thermal engineering r.s.khurmi s chand & co basic and applied thermodynamics p.k.nag tata mcgraw-hill t2 niranjan.murthy sapna publications t3 i c engines v. ganeshan tata mcgraw-hill r. c. patel and karmchandani.c .j. - elements of heat engines – acharya r1 book r2 lakshminarayana - thermal engineering vol.-i r3 n.k.giri - automobile mechanics – khanna publishers r. rudra moorthy and k. mayil swamy - heat and mass transfer-pearson r4 education india r5 j g giles-vehicle operations and testing r6 v. ganesan - internal combustion engines – tata mcgraw-hill s. p. sen - internal combustion engines theory and practice – khanna r7 publishers r8 mathur and sharma - internal combustion engines – dhanpat rai r9 v.l. malaev - ic engines – mcgraw-hill r10 p.m. heldt - ic engines r11 ramalingam - thermal engineering - new age international b b ghosh - a text book of engineering thermodynamics – vikas r12 publishing online resources: sl. no website link https://www.toppr.com/guides/chemistry/thermodynamics/applications-of- 1 thermodynamics/ 2 https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch21/chemical.php 3 https://en.wikipedia.org/wiki/thermodynamic_process http://www.thermodynamicsheatengines.com/heatenginesvol%202%20chapt 4 er%207%20rs.pdf https://www.academia.edu/15057433/chapter_8_testing_of_i.c.engines,h 5 ttps://www.mechanicalbooster.com/2015/08/difference-between-otto-cycle- and-diesel-cycle.html 6 https://engineeringinsider.org/heat-transfer-conduction-convection-radiation/ https://www.brighthubengineering.com/hvac/63725-effects-of-a-multi-stage- 7 air-compressor/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What multistage air compressors. Explain the merits 4 Understanding and demerits of multistage compression. Text / Reference: T/R Book Title/Author T1 Thermal Engineering R.S.Khurmi S Chand & Co Basic and applied thermodynamics P.K.Nag Tata MCgraw-Hill T2 Niranjan.Murthy Sapna Publications T3 I C Engines V. Ganeshan Tata MCgraw-Hill R. C. Patel and Karmchandani.C .J. - Elements of Heat engines – Acharya R1 book R2 Lakshminarayana - Thermal engineering Vol.-I R3 N.K.Giri - Automobile mechanics – Khanna publishers R. Rudra Moorthy and K. Mayil Swamy - Heat and Mass transfer- pearson R4 education india R5 J G Giles-Vehicle operations and testing R6 V. Ganesan - Internal combustion engines – Tata McGraw-Hill S. P. Sen - Internal combustion engines theory and practice – Khanna R7 publishers R8 Mathur and Sharma - Internal combustion engines – Dhanpat Rai R9 V.L. Malaev - IC Engines – McGraw-Hill R10 P.M. Heldt - IC Engines R11 Ramalingam - Thermal engineering - New age international B B Ghosh - A Text book of Engineering Thermodynamics – Vikas R12 publishing Online Resources: Sl. No Website Link https://www.toppr.com/guides/chemistry/thermodynamics/applications-of- 1 thermodynamics/ 2 https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch21/chemical.php 3 https://en.wikipedia.org/wiki/Thermodynamic_process http://www.thermodynamicsheatengines.com/HeatEnginesVol%202%20Chapt 4 er%207%20RS.pdf https://www.academia.edu/15057433/CHAPTER_8_Testing_of_I.C.Engines,h 5 ttps://www.mechanicalbooster.com/2015/08/difference-between-otto-cycle- and-diesel-cycle.html 6 https://engineeringinsider.org/heat-transfer-conduction-convection-radiation/ https://www.brighthubengineering.com/hvac/63725-effects-of-a-multi-stage- 7 air-compressor/ means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ multistage air compressors. Explain the merits 4 Understanding and demerits of multistage compression. Text / Reference: T/R Book Title/Author T1 Thermal Engineering R.S.Khurmi S Chand & Co Basic and applied thermodynamics P.K.Nag Tata MCgraw-Hill T2 Niranjan.Murthy Sapna Publications T3 I C Engines V. Ganeshan Tata MCgraw-Hill R. C. Patel and Karmchandani.C .J. - Elements of Heat engines – Acharya R1 book R2 Lakshminarayana - Thermal engineering Vol.-I R3 N.K.Giri - Automobile mechanics – Khanna publishers R. Rudra Moorthy and K. Mayil Swamy - Heat and Mass transfer- pearson R4 education india R5 J G Giles-Vehicle operations and testing R6 V. Ganesan - Internal combustion engines – Tata McGraw-Hill S. P. Sen - Internal combustion engines theory and practice – Khanna R7 publishers R8 Mathur and Sharma - Internal combustion engines – Dhanpat Rai R9 V.L. Malaev - IC Engines – McGraw-Hill R10 P.M. Heldt - IC Engines R11 Ramalingam - Thermal engineering - New age international B B Ghosh - A Text book of Engineering Thermodynamics – Vikas R12 publishing Online Resources: Sl. No Website Link

https://www.toppr.com/guides/chemistry/thermodynamics/applications-of- 1 thermodynamics/ 2 https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch21/chemical.php 3 https://en.wikipedia.org/wiki/Thermodynamic_process http://www.thermodynamicsheatengines.com/HeatEnginesVol%202%20Chapt 4 er%207%20RS.pdf https://www.academia.edu/15057433/CHAPTER_8_Testing_of_I.C.Engines,h 5 ttps://www.mechanicalbooster.com/2015/08/difference-between-otto-cycle- and-diesel-cycle.html 6 https://engineeringinsider.org/heat-transfer-conduction-convection-radiation/ https://www.brighthubengineering.com/hvac/63725-effects-of-a-multi-stage- 7 air-compressor/ □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
compressors.
             Explain the different Modes of heat transfer.
   M4.01     Identify the fields of application of heat        4
Applying
             transfer.
             Compare three modes of heat transfer and
                                                               1
Understanding
   M4.02     their rate equations.
             Illustrate two types of convective mode of
   M4.03                                                        2
Understanding
             heat transfer- natural and forced.
             Define air compressor, explain uses of
   M4.04                                                        3
Understanding
             compressed air and classify air compressors.
             Illustrate the working of single stage and
   M4.05     multistage air compressors. Explain the merits     4
Understanding
             and demerits of multistage compression.
             Series Test -II                                    1

Text / Reference:

   T/R                                    Book Title/Author
    T1      Thermal Engineering R.S.Khurmi S Chand & Co
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

## Concept and formula bank

### Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

**Define or explain Define basic terms of thermodynamics 3 Remembering Explain First, Second, Third and Zeroth Laws M10.2 1 Understanding of thermodynamics. Summarize the basic concepts of entropy,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Define basic terms of thermodynamics 3 Remembering Explain First, Second, Third and Zeroth Laws M10.2 1 Understanding of thermodynamics. Summarize the basic concepts of entropy,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 2 Understanding enthalpy and internal energy. Make use of various types of thermodynamic. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding enthalpy and internal energy. Make use of various types of thermodynamic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 6 Applying processes. Solve problems on different types of. (3 marks)**

**Answer:** Begin with the standard definition or identity of *6 Applying processes. Solve problems on different types of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 3 Applying thermodynamic processes. Fundamentals of thermodynamics and thermodynamic processes Classification of thermodynamics, Application areas of thermodynamics, Dimensional homogeneity, Basics of thermodynamics - thermodynamic system, surroundings, boundary, types of system, properties of a system, state, thermodynamic equilibrium, Ideal gas equation of state - Derivation, pressure and its units, temperature and its units, Zeroth law of thermodynamics, Specific heat at constant pressure and constant volume, relationship between specific heats and gas constant, Internal Energy, flow energy and moving boundary work, First law of thermodynamics - Enthalpy, Second law of**

**thermodynamics, Entropy-Third law of thermodynamics. Processes and cycle. Pressure volume diagram, Temperature entropy diagram, Types of processes - Derivations on PVT relationship, heat transfer, work transfer, change in entropy, enthalpy and internal energy of isochoric, isobaric, isothermal and adiabatic processes. Simple problems to find change in internal energy, heat transfer, work transfer etc.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Applying thermodynamic processes. Fundamentals of thermodynamics and thermodynamic processes Classification of thermodynamics, Application areas of thermodynamics, Dimensional homogeneity, Basics of thermodynamics - thermodynamic system, surroundings, boundary, types of system, properties of a system, state, thermodynamic equilibrium, Ideal gas equation of state - Derivation, pressure and its units, temperature and its units, Zeroth law of thermodynamics, Specific heat at constant pressure and constant volume, relationship between specific heats and gas constant, Internal Energy, flow energy and moving boundary work, First law of thermodynamics - Enthalpy, Second law of thermodynamics, Entropy-Third law of thermodynamics. Processes and cycle. Pressure volume diagram, Temperature entropy diagram, Types of processes - Derivations on PVT relationship, heat transfer, work transfer, change in entropy, enthalpy and internal energy of isochoric, isobaric, isothermal and adiabatic processes. Simple problems to find change in internal energy, heat transfer, work transfer etc.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 2

**Define or explain thermodynamic cycles. Classify and compare 2 Understanding, thermodynamic cycles. Define available energy, theoretical thermal. (3 marks)**

**Answer:** Begin with the standard definition or identity of *thermodynamic cycles. Classify and compare 2 Understanding, thermodynamic cycles. Define available energy, theoretical thermal*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 2 Remembering, efficiency / air standard efficiency. Summarize the theoretical thermal efficiency. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Remembering, efficiency / air standard efficiency. Summarize the theoretical thermal efficiency*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 2 Understanding of Carnot cycle.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding of Carnot cycle.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Solve simple problems on Carnot cycle. 2 Applying Explain theoretical thermal efficiency of Otto. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Solve simple problems on Carnot cycle. 2 Applying Explain theoretical thermal efficiency of Otto*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 3

**Define or explain efficiency, brake thermal efficiency, 6 Understanding volumetric efficiency, Mechanical efficiency, Mean effective pressures, and specific fuel consumption. Solve the problems related to testing of. (3 marks)**

**Answer:** Begin with the standard definition or identity of *efficiency, brake thermal efficiency, 6 Understanding volumetric efficiency, Mechanical efficiency, Mean effective pressures, and specific fuel consumption. Solve the problems related to testing of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 4 Applying Internal combustion engines. Summarize how Morse test is conducted in. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Applying Internal combustion engines. Summarize how Morse test is conducted in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 1 Understanding multi cylinder engines.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *1 Understanding multi cylinder engines.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Explain how heat balance sheets are prepared. 1 Understanding Solve the problems on Morse test and Heat. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain how heat balance sheets are prepared. 1 Understanding Solve the problems on Morse test and Heat*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 4

**Define or explain Identify the fields of application of heat 4 Applying transfer. Compare three modes of heat transfer and. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Identify the fields of application of heat 4 Applying transfer. Compare three modes of heat transfer and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain their rate equations. Illustrate two types of convective mode of. (3 marks)**

**Answer:** Begin with the standard definition or identity of *their rate equations. Illustrate two types of convective mode of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 2 Understanding heat transfer- natural and forced. Define air compressor, explain uses of. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding heat transfer- natural and forced. Define air compressor, explain uses of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 3 Understanding compressed air and classify air compressors. Illustrate the working of single stage and. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding compressed air and classify air compressors. Illustrate the working of single stage and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **Define basic terms of thermodynamics 3 Remembering Explain First, Second, Third and Zeroth Laws M10.2 1 Understanding of thermodynamics. Summarize the basic concepts of entropy,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **thermodynamic cycles. Classify and compare 2 Understanding, thermodynamic cycles. Define available energy, theoretical thermal**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **efficiency, brake thermal efficiency, 6 Understanding volumetric efficiency, Mechanical efficiency, Mean effective pressures, and specific fuel consumption. Solve the problems related to testing of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **Identify the fields of application of heat 4 Applying transfer. Compare three modes of heat transfer and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

## Glossary

**Study glossary**

| Term | Meaning in this lesson |
| --- | --- |
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module | An official syllabus unit with defined topics and hours. |
| CIE | Continuous Internal Evaluation. |
| ESE | End Semester Examination. |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

SOURCE-SUPPORTED REFERENCES

# References and web resources

## References listed in the official PDF

References are included in the official source PDF.

## Web resources listed in the official PDF


- Sl. No Website Link
- https://www.toppr.com/guides/chemistry/thermodynamics/applications-of-thermodynamics/
- https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch21/chemical.php
- https://en.wikipedia.org/wiki/Thermodynamic_process
- http://www.thermodynamicsheatengines.com/HeatEnginesVol%202%20Chapter%207%20RS.pdf
- https://www.academia.edu/15057433/CHAPTER_8_Testing_of_I.C.Engines,h
- ttps://www.mechanicalbooster.com/2015/08/difference-between-otto-cycle- and-diesel-cycle.html
- https://engineeringinsider.org/heat-transfer-conduction-convection-radiation/
- https://www.brighthubengineering.com/hvac/63725-effects-of-a-multi-stage- air-compressor/


IN-PAGE SEARCH

# Search this lesson